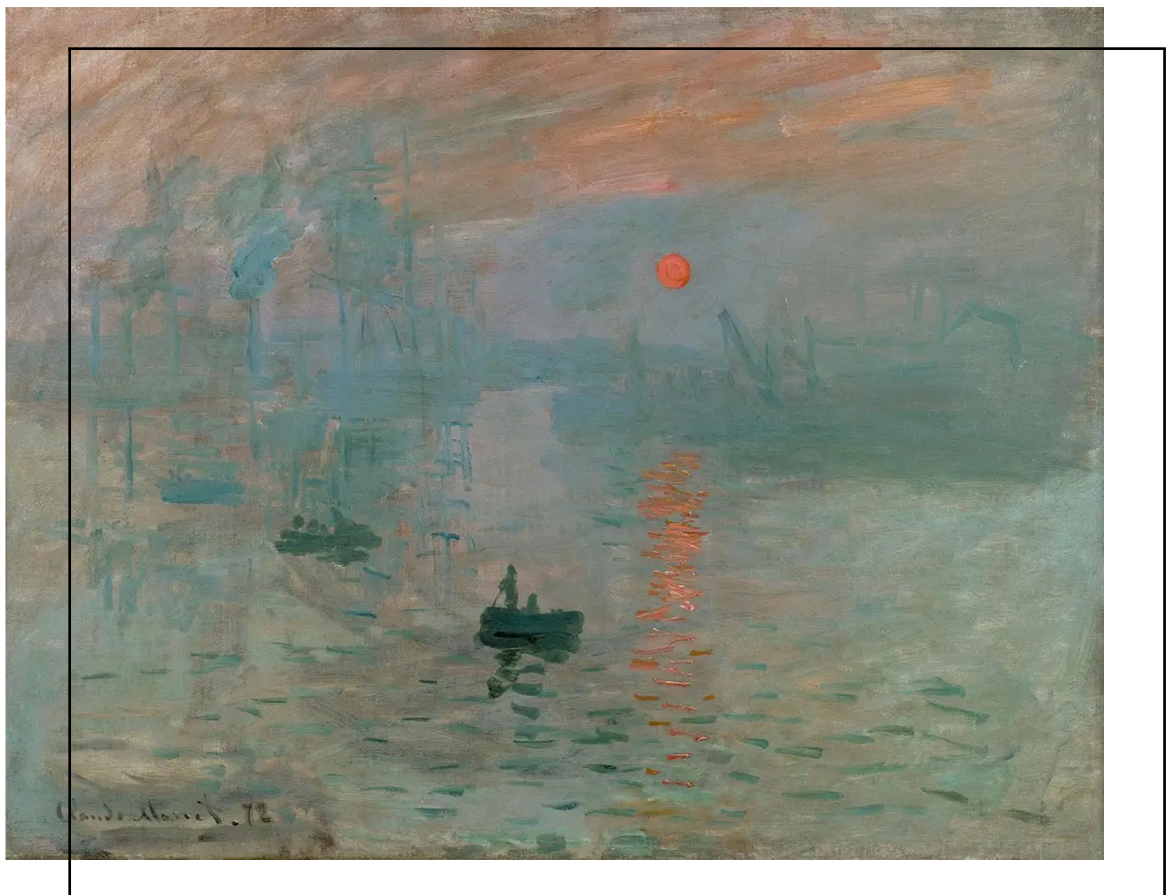


01

Homework - General Relativity

Newtonian Gravity

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Q. 01

Gravitational Field: a sphere, pt. 1

Find the gravitational field inside and outside a sphere of radius R , which carries a uniform volume mass density ρ .

To find the gravitational field inside the sphere, we can use Gauss's law, which states that the gravitational flux through a closed surface is proportional to the enclosed mass. So, we can consider a spherical Gaussian surface of radius r (where $r < R$) centered at the center of the sphere. The mass enclosed by this Gaussian surface is given by:

$$M_{\text{enclosed}} = \int_V \rho(\mathbf{r}') d^3\mathbf{r}' = \rho \int_0^r \int_0^\pi \int_0^{2\pi} (r')^2 \sin \theta' d\phi' d\theta' dr' = \frac{4}{3}\pi r^3 \rho$$

The gravitational field at a distance r from the center of the sphere can be found using Gauss's law too:

$$\oint_S \mathbf{g}(\mathbf{r}) \cdot \hat{\mathbf{n}} dA = -4\pi G M_{\text{enclosed}}$$

Since the gravitational field is radial and has the same magnitude at every point on the Gaussian surface, we can simplify this to:

$$g(r)4\pi r^2 = -4\pi G \left(\frac{4}{3}\pi r^3 \rho \right)$$

This gives us the gravitational field inside the sphere:

$$\mathbf{g}(\mathbf{r}) = -\frac{4}{3}\pi G \rho r \hat{\mathbf{r}}, \quad \text{for } r < R \quad (1.1)$$

To find the gravitational field outside the sphere, we can use the same argument, but now the Gaussian surface is a sphere of radius r (where $r > R$) that encloses the entire mass of the sphere. Therefore, the enclosed mass is:

$$M_{\text{enclosed}} = \frac{4}{3}\pi R^3 \rho$$

Using the same application of Gauss's law to find the gravitational field, we have:

$$g(r)4\pi r^2 = -4\pi G M_{\text{enclosed}} = -4\pi G \left(\frac{4}{3}\pi R^3 \rho \right)$$



This gives us the gravitational field outside the sphere:

$$\mathbf{g}(\mathbf{r}) = -\frac{4}{3}\pi G\rho\frac{R^3}{r^2}\hat{\mathbf{r}}, \quad \text{for } r > R \quad (1.2)$$





Q. 02 Gravitational Field: a spherical shell, pt. 1

Find the gravitational field inside and outside a spherical shell of radius R , which carries a uniform surface mass density σ .

The gravitational field inside the spherical shell is zero, since there is no mass enclosed by any Gaussian surface inside the shell. Therefore, we have:

$$\mathbf{g}(\mathbf{r}) = \mathbf{0}, \quad \text{for } r < R \quad (2.1)$$

To find the gravitational field outside the spherical shell, we can use Gauss's law again. The mass enclosed by a Gaussian surface of radius r (where $r > R$) is given by:

$$M_{\text{enclosed}} = \int_S \sigma(\mathbf{r}') d^2\mathbf{r}' = \sigma \int_0^\pi \int_0^{2\pi} R^2 \sin \theta' d\phi' d\theta' = 4\pi R^2 \sigma$$

We can use again that:

$$\oint_S \mathbf{g}(\mathbf{r}) \cdot \hat{\mathbf{n}} dA = -4\pi G M_{\text{enclosed}}$$

Since the gravitational field is radial and has the same magnitude at every point on the Gaussian surface, we can simplify this to:

$$g(r)4\pi r^2 = -4\pi G M_{\text{enclosed}} = -4\pi G (4\pi R^2 \sigma)$$

This gives us the gravitational field outside the spherical shell:

$$\mathbf{g}(\mathbf{r}) = -\frac{4\pi G R^2 \sigma}{r^2} \hat{\mathbf{r}}, \quad \text{for } r > R \quad (2.2)$$





Q. 03

Gravitational potential: a straight wire

Find the gravitational potential a distance s from an infinitely long straight wire that carries a uniform line charge λ . Compute the gradient of your potential and find the corresponding gravitational field.

To find the gravitational potential at a distance s from an infinitely long straight wire, we can use:

$$U(\mathbf{s}) = G \int \frac{dm}{\|\mathbf{s}\|}$$

Since the wire is infinitely long, we can consider a small segment of the wire of length dz , which has a mass $dm = \lambda dz$. The distance from this segment to the point where we want to calculate the potential is given by $r = \sqrt{s^2 + z^2}$. Taking a segment with length $2L$ centered at $z = 0$, we can write:

$$U(s) = G \int_{-L}^L \frac{\lambda}{\sqrt{s^2 + z^2}} dz = G\lambda \int_{-L}^L \frac{1}{\sqrt{s^2 + z^2}} dz$$

The indefinite integral has a known solution:

$$\int \frac{1}{\sqrt{s^2 + z^2}} dz = \ln(z + \sqrt{s^2 + z^2}) + C$$

So, the potential can be calculated as:

$$U(s) = G\lambda \left[\ln(L + \sqrt{s^2 + L^2}) - \ln(-L + \sqrt{s^2 + L^2}) \right] = G\lambda \ln \left(\frac{L + \sqrt{s^2 + L^2}}{-L + \sqrt{s^2 + L^2}} \right)$$

The argument of the logarithm can be simplified as:

$$\frac{L + \sqrt{s^2 + L^2}}{-L + \sqrt{s^2 + L^2}} = \frac{L + \sqrt{s^2 + L^2}}{-L + \sqrt{s^2 + L^2}} \cdot \frac{L + \sqrt{s^2 + L^2}}{L + \sqrt{s^2 + L^2}} = \frac{(L + \sqrt{s^2 + L^2})^2}{s^2}$$

For large L , we can approximate $\sqrt{s^2 + L^2} \approx L$, which gives us:

$$U(s) = 2G\lambda \ln \left(\frac{2L}{s} \right) = 2G\lambda \ln(2L) - 2G\lambda \ln(s)$$

The first term diverges as $L \rightarrow \infty$, but does not depend on s , so it has no physical effect to the gravitational field. This can be absorbed into a reference potential, i.e., the difference:

$$U(s) - U(s_0) = -2G\lambda \ln(s) + 2G\lambda \ln(s_0) = -2G\lambda \ln \left(\frac{s}{s_0} \right)$$



Choosing s_0 such that $U(s_0) = 0$, we can write the potential as:

$$U(s) = -2G\lambda \ln\left(\frac{s}{s_0}\right) \quad (3.1)$$

To find the gravitational field, we can compute the gradient of the potential:

$$\mathbf{g}(\mathbf{s}) = \nabla U(s) = -2G\lambda \nabla \left[\ln\left(\frac{s}{s_0}\right) \right] \hat{\mathbf{s}} = -2G\lambda \frac{d}{ds} \left[\ln\left(\frac{s}{s_0}\right) \right] \hat{\mathbf{s}} = -2G\lambda \frac{1}{s} \hat{\mathbf{s}}$$

Concluding that:

$$\mathbf{g}(\mathbf{s}) = -\frac{2G\lambda}{s} \hat{\mathbf{s}} \quad (3.2)$$





Q. 04 Gravitational potential: a hollow sphere

The gravitational potential $U(\theta)$ is specified on the surface of a hollow sphere of radius R . Find the potential inside the sphere.

Knowing that $\nabla \cdot \mathbf{g} = -4\pi G\rho$, we can use that $\mathbf{g} = \nabla U$ to obtain the Poisson equation for the gravitational potential $\nabla^2 U = -4\pi G\rho$. In the case of a hollow sphere, we have $\rho = 0$ everywhere, so the Poisson equation reduces to the Laplace equation $\nabla^2 U = 0$. The general solution to the Laplace equation in spherical coordinates can be written as:

$$U(r, \theta) = \sum_{\ell=0}^{\infty} \left[A_{\ell} r^{\ell} + \frac{B_{\ell}}{r^{\ell+1}} \right] P_{\ell}(\cos \theta)$$

Inside the sphere, the solution must be finite at $r = 0$, which implies that $B_{\ell} = 0$ for all ℓ . Therefore, the potential inside the sphere can be written as:

$$U_{\text{inside}}(r, \theta) = \sum_{\ell=0}^{\infty} A_{\ell} r^{\ell} P_{\ell}(\cos \theta)$$

On the surface of the sphere, we have $U(R, \theta) = U(\theta)$, which gives us:

$$U(\theta) = \sum_{\ell=0}^{\infty} A_{\ell} R^{\ell} P_{\ell}(\cos \theta)$$

Multiplying both sides by $P_{\ell'}(\cos \theta)$ and integrating over $d \cos \theta$ from -1 to 1 :

$$\int_{-1}^1 U(\theta) P_{\ell'}(\cos \theta) d \cos \theta = \sum_{\ell=0}^{\infty} A_{\ell} R^{\ell} \int_{-1}^1 P_{\ell}(\cos \theta) P_{\ell'}(\cos \theta) d \cos \theta$$

Using the orthogonality of the Legendre polynomials, we have for the integral on the right-hand side the result $\frac{2}{2\ell'+1} \delta_{\ell\ell'}$, therefore:

$$\int_{-1}^1 U(\theta) P_{\ell'}(\cos \theta) d \cos \theta = A_{\ell'} R^{\ell'} \frac{2}{2\ell' + 1}$$

implying that:

$$A_{\ell} = \frac{2\ell + 1}{2R^{\ell}} \int_{-1}^1 U(\theta) P_{\ell}(\cos \theta) d \cos \theta \quad (4.1)$$



Thus, the potential inside the sphere can be expressed as:

$$U_{\text{inside}}(r, \theta) = \sum_{\ell=0}^{\infty} \frac{2\ell+1}{2R^{\ell}} r^{\ell} P_{\ell}(\cos \theta) \int_{-1}^1 U(\theta) P_{\ell}(\cos \theta) d \cos \theta \quad (4.2)$$





Q. 05

Gravitational potential: a sphere, pt. 2

The potential at the surface of a sphere of radius R is given by $U(\theta) = k \cos(3\theta)$, where k is a constant. Find the potential inside and outside the sphere, as well as the surface mass density $\sigma(\theta)$ on the sphere (assume there is no mass inside or outside the sphere).

By the Question 04, we know that we can write $U(r, \theta)$ in terms of the Legendre polynomials, therefore, we can express $U(\theta) = k \cos(3\theta)$ as a linear combination of some Legendre polynomials. Before we do that, the $\cos(3\theta)$ can be expanded as:

$$\cos(3\theta) = -3 \cos \theta + 4 \cos^3 \theta$$

This means that we need Legendre polynomials of 1st and 3rd order in $\cos \theta$. The first few Legendre polynomials are:

$$P_0(\cos \theta) = 1$$

$$P_1(\cos \theta) = \cos \theta$$

$$P_2(\cos \theta) = \frac{1}{2}(3 \cos^2 \theta - 1)$$

$$P_3(\cos \theta) = \frac{1}{2}(5 \cos^3 \theta - 3 \cos \theta)$$

In these cases, we can note that only $P_1(\cos \theta)$ and $P_3(\cos \theta)$ are needed to express $\cos(3\theta)$, therefore, we can write:

$$\cos(3\theta) = AP_1(\cos \theta) + BP_3(\cos \theta)$$

where A and B are constants to be determined. Substituting the expressions for $P_1(\cos \theta)$ and $P_3(\cos \theta)$, we have:

$$\begin{aligned} \cos(3\theta) &= A \cos \theta + B \left[\frac{1}{2}(5 \cos^3 \theta - 3 \cos \theta) \right] \\ &= \left(A - \frac{3}{2}B \right) \cos \theta + \frac{5}{2}B \cos^3 \theta \end{aligned}$$

Comparing the coefficients of the two expressions for $\cos(3\theta)$, we obtain:

$$\frac{5}{2}B = 4 \implies B = \frac{8}{5} \quad \text{and} \quad A - \frac{3}{2}B = -3 \implies A = -\frac{12}{5}$$

Thus, we have:

$$U(\theta) = \frac{2k}{5} [-6P_1(\cos \theta) + 4P_3(\cos \theta)]$$



Knowing that $U(\theta)$ depends only on $P_1(\cos \theta)$ and $P_3(\cos \theta)$, the potential inside the sphere will be expressed only for $\ell = 1$ and $\ell = 3$, because of the orthogonality of the Legendre polynomials. Therefore, we can write:

$$U_{\text{inside}}(r, \theta) = A_1 r P_1(\cos \theta) + A_3 r^3 P_3(\cos \theta)$$

where A_ℓ is given by (4.1). Calculating A_1 :

$$\begin{aligned} A_1 &= \frac{3}{2R} \int_{-1}^1 U(\theta) P_1(\cos \theta) d \cos \theta \\ &= \frac{3}{2R} \int_{-1}^1 \frac{2k}{5} [-6P_1(\cos \theta) + 4P_3(\cos \theta)] P_1(\cos \theta) d \cos \theta \\ &= -\frac{36k}{10R} \int_{-1}^1 P_1^2(\cos \theta) d \cos \theta \end{aligned}$$

Using that:

$$\int_{-1}^1 P_\ell^2(\cos \theta) d \cos \theta = \frac{2}{2\ell + 1}$$

we have:

$$A_1 = -\frac{36k}{10R} \cdot \frac{2}{3} = -\frac{12k}{5R}$$

For A_3 , we have:

$$\begin{aligned} A_3 &= \frac{7}{2R^3} \int_{-1}^1 U(\theta) P_3(\cos \theta) d \cos \theta \\ &= \frac{7}{2R^3} \int_{-1}^1 \frac{2k}{5} [-6P_1(\cos \theta) + 4P_3(\cos \theta)] P_3(\cos \theta) d \cos \theta \\ &= \frac{56k}{10R^3} \int_{-1}^1 P_3^2(\cos \theta) d \cos \theta \\ &= \frac{56k}{10R^3} \cdot \frac{2}{7} = \frac{8k}{5R^3} \end{aligned}$$

Thus, the potential inside the sphere can be expressed as:

$$\begin{aligned} U_{\text{inside}}(r, \theta) &= -\frac{12k}{5R} r P_1(\cos \theta) + \frac{8k}{5R^3} r^3 P_3(\cos \theta) \\ &= -\frac{12k}{5R} r \cos \theta + \frac{8k}{5R^3} r^3 \cdot \frac{1}{2} (5 \cos^3 \theta - 3 \cos \theta) \\ &= \frac{4k}{5R} r \cos \theta \left[-3 + \frac{r^2}{2R^2} (5 \cos^2 \theta - 3) \right] \end{aligned}$$

Concluding that the potential inside the sphere is given by:



$$U_{\text{inside}}(r, \theta) = \frac{4k}{5R} r \cos \theta \left[-3 + \frac{r^2}{2R^2} (5 \cos^2 \theta - 3) \right] \quad (5.1)$$

Outside the sphere, we can use the general solution to the Laplace equation to determine the potential. In this case, the solution must vanish at infinity, which implies that $A_\ell = 0$ for all ℓ . Therefore, the potential outside the sphere can be written as:

$$U_{\text{outside}}(r, \theta) = \sum_{\ell=0}^{\infty} \frac{B_\ell}{r^{\ell+1}} P_\ell(\cos \theta)$$

On the surface of the sphere, we have $U(R, \theta) = U(\theta)$, which gives us:

$$U(\theta) = \sum_{\ell=0}^{\infty} \frac{B_\ell}{R^{\ell+1}} P_\ell(\cos \theta)$$

Multiplying both sides by $P_{\ell'}(\cos \theta)$ and integrating over $d \cos \theta$ from -1 to 1 :

$$\begin{aligned} \int_{-1}^1 U(\theta) P_{\ell'}(\cos \theta) d \cos \theta &= \sum_{\ell=0}^{\infty} \frac{B_\ell}{R^{\ell+1}} \int_{-1}^1 P_\ell(\cos \theta) P_{\ell'}(\cos \theta) d \cos \theta \\ &= \frac{2B_{\ell'}}{(2\ell' + 1)R^{\ell'+1}} \end{aligned}$$

implying that:

$$B_\ell = \frac{2\ell + 1}{2} R^{\ell+1} \int_{-1}^1 U(\theta) P_\ell(\cos \theta) d \cos \theta$$

By orthogonality of the Legendre polynomials, only B_1 and B_3 will be non-zero, therefore:

$$\begin{aligned} B_1 &= \frac{3}{2} R^2 \int_{-1}^1 U(\theta) P_1(\cos \theta) d \cos \theta \\ &= \frac{3}{2} R^2 \int_{-1}^1 \frac{2k}{5} [-6P_1(\cos \theta) + 4P_3(\cos \theta)] P_1(\cos \theta) d \cos \theta \\ &= -\frac{36k}{10} R^2 \int_{-1}^1 P_1^2(\cos \theta) d \cos \theta \\ &= -\frac{36k}{10} R^2 \cdot \frac{2}{3} \\ &= -\frac{12k}{5} R^2 \end{aligned}$$

and



$$\begin{aligned}
B_3 &= \frac{7}{2} R^4 \int_{-1}^1 U(\theta) P_3(\cos \theta) d \cos \theta \\
&= \frac{7}{2} R^4 \int_{-1}^1 \frac{2k}{5} [-6P_1(\cos \theta) + 4P_3(\cos \theta)] P_3(\cos \theta) d \cos \theta \\
&= \frac{56k}{10} R^4 \int_{-1}^1 P_3^2(\cos \theta) d \cos \theta \\
&= \frac{56k}{10} R^4 \cdot \frac{2}{7} \\
&= \frac{8k}{5} R^4
\end{aligned}$$

Thus, the potential outside the sphere can be expressed as:

$$\begin{aligned}
U_{\text{outside}}(r, \theta) &= -\frac{12k}{5} \frac{R^2}{r^2} P_1(\cos \theta) + \frac{8k}{5} \frac{R^4}{r^4} P_3(\cos \theta) \\
&= -\frac{12k}{5} \frac{R^2}{r^2} \cos \theta + \frac{8k}{5} \frac{R^4}{r^4} \cdot \frac{1}{2} (5 \cos^3 \theta - 3 \cos \theta) \\
&= \frac{4k}{5} \frac{R^2}{r^2} \cos \theta \left[-3 + \frac{R^2}{2r^2} (5 \cos^2 \theta - 3) \right]
\end{aligned}$$

Concluding that the potential outside the sphere is given by:

$$U_{\text{outside}}(r, \theta) = \frac{4k}{5} \frac{R^2}{r^2} \cos \theta \left[-3 + \frac{R^2}{2r^2} (5 \cos^2 \theta - 3) \right] \quad (5.2)$$

Because there is no mass inside or outside the sphere, the surface mass density $\sigma(\theta)$ can be obtained from the discontinuity of the normal component of the gravitational field across the surface of the sphere:

$$g_{\text{outside}}^{(r)} - g_{\text{inside}}^{(r)} = -4\pi G \sigma(\theta) \implies \sigma(\theta) = -\frac{1}{4\pi G} \left[\frac{\partial U_{\text{outside}}}{\partial r} - \frac{\partial U_{\text{inside}}}{\partial r} \right]_{r=R}$$

For the derivative of U_{inside} with respect to r , we have:

$$\frac{\partial U_{\text{inside}}}{\partial r} = -\frac{12k}{5R} P_1(\cos \theta) + \frac{24k}{5R^3} r^2 P_3(\cos \theta)$$

and for the derivative of U_{outside} with respect to r , we have:

$$\frac{\partial U_{\text{outside}}}{\partial r} = \frac{24k}{5} \frac{R^2}{r^3} P_1(\cos \theta) - \frac{32k}{5} \frac{R^4}{r^5} P_3(\cos \theta)$$



Thus:

$$\begin{aligned}\left[\frac{\partial U_{\text{outside}}}{\partial r} - \frac{\partial U_{\text{inside}}}{\partial r}\right]_{r=R} &= \frac{12k}{5R}P_1(\cos\theta) - \frac{8k}{5R}P_3(\cos\theta) \\ &= \frac{4k}{5R}[3P_1(\cos\theta) - 2P_3(\cos\theta)] \\ &= \frac{4k}{5R}[3\cos\theta - (5\cos^3\theta - 3\cos\theta)] \\ &= \frac{4k}{5R}(6\cos\theta - 5\cos^3\theta)\end{aligned}$$

Concluding that the surface mass density on the sphere is given by:

$$\sigma(\theta) = -\frac{k}{5\pi GR}(6\cos\theta - 5\cos^3\theta) \quad (5.3)$$





Q. 06

Gravitational potential: a spherical shell, pt. 2

Derive the exact gravitational potential for a spherical shell of radius R which carries a surface mass density $\sigma(\theta) = k \cos(\theta)$, where k is a constant. Calculate also the quadrupole moment of this mass distribution.

Knowing the form of $\sigma(\theta)$, it can be written as $\sigma(\theta) = k P_1(\cos \theta)$. The general solutions of Laplace's equation give us:

$$U_{\text{inside}}(r, \theta) = \sum_{\ell=0}^{\infty} A_{\ell} r^{\ell} P_{\ell}(\cos \theta) \quad \text{and} \quad U_{\text{outside}}(r, \theta) = \sum_{\ell=0}^{\infty} \frac{B_{\ell}}{r^{\ell+1}} P_{\ell}(\cos \theta)$$

By the discontinuity of the normal component of the gravitational field across the shell, we have:

$$\begin{aligned} \sigma(\theta) &= \frac{1}{4\pi G} \left[\frac{\partial U_{\text{outside}}}{\partial r} - \frac{\partial U_{\text{inside}}}{\partial r} \right]_{r=R} \\ &= \frac{1}{4\pi G} \left[\sum_{\ell=0}^{\infty} A_{\ell} \ell R^{\ell-1} P_{\ell}(\cos \theta) + \sum_{\ell=0}^{\infty} (\ell+1) \frac{B_{\ell}}{R^{\ell+2}} P_{\ell}(\cos \theta) \right] \end{aligned}$$

If $\sigma(\theta)$ depends only on $P_1(\cos \theta)$, then all terms in the above expression must vanish except for the one with $\ell = 1$, therefore:

$$\begin{aligned} \sigma(\theta) &= \frac{1}{4\pi G} \left[A_1 R^0 P_1(\cos \theta) + \frac{2B_1}{R^3} P_1(\cos \theta) \right] \\ &= \frac{1}{4\pi G} \left[A_1 + \frac{2B_1}{R^3} \right] P_1(\cos \theta) \end{aligned}$$

We know that $U_{\text{inside}}(R, \theta) = U_{\text{outside}}(R, \theta)$, which implies that $A_1 R P_1(\cos \theta) = \frac{B_1}{R^2} P_1(\cos \theta)$, therefore $B_1 = A_1 R^3$. Substituting this result in the above expression for $\sigma(\theta)$, we have:

$$\begin{aligned} \sigma(\theta) &= \frac{1}{4\pi G} [A_1 + 2A_1] P_1(\cos \theta) \\ &= \frac{3A_1}{4\pi G} P_1(\cos \theta) \end{aligned}$$

Thus:

$$\frac{3A_1}{4\pi G} = k \implies A_1 = \frac{4\pi G}{3} k \implies B_1 = \frac{4\pi G}{3} k R^3$$

Therefore the potential inside the sphere can be expressed as:



$$U_{\text{inside}}(r, \theta) = 4\pi G \frac{kr}{3} \cos \theta \quad (6.1)$$

and the potential outside the sphere can be expressed as:

$$U_{\text{outside}}(r, \theta) = 4\pi G \frac{kR^3}{3r^2} \cos \theta \quad (6.2)$$

We can see that inside the sphere the potential is proportional to r , which give us a uniform gravitational field, while outside the sphere the potential is proportional to $\cos \theta / r^2$, which give us a dipole term in the multipole expansion of $U(\mathbf{r})$. This implies that the quadrupole moment of this mass distribution is zero, which can be verified by direct calculation:

$$\begin{aligned} Q_{ij} &= \int \sigma(\theta') (3r'_i r'_j - r'^2 \delta_{ij}) d^2 r' \\ &= k \int \cos \theta' (3r'_i r'_j - r'^2 \delta_{ij}) d^2 r' \\ &= kR^2 (3r'_i r'_j - \delta_{ij}) \int_0^\pi \int_0^{2\pi} \cos \theta' \sin \theta' d\phi' d\theta' \\ &= 2\pi kR^2 (3r'_i r'_j - \delta_{ij}) \int_0^\pi \cos \theta' \sin \theta' d\theta' \\ &= 0 (\text{by ortogonality}) \end{aligned}$$

Concluding that the quadrupole moment of this mass distribution is zero:

$$Q_{ij} = 0 \quad (6.3)$$

